Test Series Code

14

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | **DROPPER** | **M. MARKS: 720**

DATE: 19/04/2026

ANSWER KEY

1.	(2)	46.	(4)	91.	(3)	136.	(1)
2.	(2)	47.	(3)	92.	(2)	137.	(1)
3.	(3)	48.	(3)	93.	(4)	138.	(1)
4.	(2)	49.	(4)	94.	(1)	139.	(3)
5.	(4)	50.	(1)	95.	(3)	140.	(2)
6.	(2)	51.	(2)	96.	(2)	141.	(2)
7.	(2)	52.	(2)	97.	(2)	142.	(2)
8.	(2)	53.	(1)	98.	(1)	143.	(1)
9.	(1)	54.	(4)	99.	(3)	144.	(1)
10.	(3)	55.	(4)	100.	(4)	145.	(1)
11.	(2)	56.	(1)	101.	(1)	146.	(2)
12.	(1)	57.	(2)	102.	(2)	147.	(3)
13.	(2)	58.	(1)	103.	(2)	148.	(4)
14.	(1)	59.	(1)	104.	(3)	149.	(3)
15.	(2)	60.	(2)	105.	(4)	150.	(3)
16.	(2)	61.	(4)	106.	(1)	151.	(4)
17.	(3)	62.	(3)	107.	(4)	152.	(2)
18.	(2)	63.	(1)	108.	(1)	153.	(2)
19.	(2)	64.	(2)	109.	(4)	154.	(2)
20.	(1)	65.	(3)	110.	(4)	155.	(2)
21.	(1)	66.	(4)	111.	(2)	156.	(4)
22.	(1)	67.	(2)	112.	(3)	157.	(1)
23.	(3)	68.	(4)	113.	(1)	158.	(4)
24.	(1)	69.	(1)	114.	(4)	159.	(2)
25.	(3)	70.	(1)	115.	(2)	160.	(1)
26.	(2)	71.	(3)	116.	(1)	161.	(3)
27.	(4)	72.	(4)	117.	(1)	162.	(3)
28.	(2)	73.	(1)	118.	(3)	163.	(3)
29.	(2)	74.	(3)	119.	(2)	164.	(3)
30.	(4)	75.	(3)	120.	(4)	165.	(2)
31.	(1)	76.	(2)	121.	(4)	166.	(3)
32.	(2)	77.	(3)	122.	(4)	167.	(2)
33.	(2)	78.	(3)	123.	(3)	168.	(3)
34.	(3)	79.	(3)	124.	(3)	169.	(2)
35.	(2)	80.	(3)	125.	(4)	170.	(4)
36.	(2)	81.	(2)	126.	(3)	171.	(4)
37.	(2)	82.	(3)	127.	(2)	172.	(3)
38.	(3)	83.	(2)	128.	(1)	173.	(2)
39.	(2)	84.	(3)	129.	(3)	174.	(4)
40.	(3)	85.	(1)	130.	(4)	175.	(2)
41.	(2)	86.	(3)	131.	(3)	176.	(2)
42.	(2)	87.	(1)	132.	(4)	177.	(3)
43.	(2)	88.	(4)	133.	(2)	178.	(3)
44.	(3)	89.	(1)	134.	(3)	179.	(1)
45.	(1)	90.	(1)	135.	(3)	180.	(3)

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Hints & Solutions

1. (2)

$$v=\frac{dx}{dt}=\alpha x^{1/2}\Rightarrow\int x^{-1/2}dx=\int\alpha dt$$

$$\Rightarrow 2x^{1/2}=\alpha t\Rightarrow x=\frac{\alpha^2t^2}{4}$$

Velocity at time t is $v=\alpha\sqrt{\frac{\alpha^2t^2}{4}}=\frac{\alpha^2t}{2}$

By Work-Energy Theorem,

$$\text{Work}=\Delta KE=\frac{1}{2}mv^2-0=\frac{1}{2}m\left(\frac{\alpha^2t}{2}\right)^2=\frac{m\alpha^4t^2}{8}$$

(NCERT Ed. 2025-26, 11th, Page No. 73)

2. (2)

When switch *S* is closed, current does not pass through bulb *C*. Hence total resistance decreases and current passing through bulbs *A* and *B* increases, therefore brightness of bulbs *A* and *B* also increases.

(NCERT Ed. 2025-26, 12th, Page No. 93)

3. (3)

Initial equilibrium extension is $x_0=\frac{mg}{k}$.

When $F = mg$ is applied, the new equilibrium position becomes $\frac{(mg+mg)}{k}=\frac{2mg}{k}$.

The mass will oscillate around this new equilibrium. Amplitude of oscillation

= New Equilibrium – Initial Position

$$=\frac{2mg}{k}-\frac{mg}{k}=\frac{mg}{k}.$$

Maximum extension from natural length = New Equilibrium + Amplitude $=\frac{2mg}{k}+\frac{mg}{k}=\frac{3mg}{k}$.

(NCERT Ed. 2025-26, 11th, Page No. 81)

4. (2)

Orbital speed $v=\sqrt{\frac{GM}{r}}$, so $v\propto\frac{1}{\sqrt{r}}$

Initial orbital radius $r_1=R+h=R+R=2R$

Final orbital radius $r_2=R+3R=4R$

Ratio $\frac{v_2}{v_1}=\sqrt{\frac{r_1}{r_2}}=\sqrt{\frac{2R}{4R}}=\frac{1}{\sqrt{2}}$

(NCERT Ed. 2025-26, 11th, Page No. 137)

5. (4)

DC voltmeter reads average value. The average of AC for complete cycle is zero.

(NCERT Ed. 2025-26, 12th, Page No. 178)

6. (2)

Horizontal distance: $x=\frac{R}{2}=\frac{v_xv_y}{g}$

Vertical distance: $y=-H_{max}=-\frac{v_y^2}{2g}$

Angular Momentum ($L=m|xv_y-yv_x|$):

$$L=m\left|\left(\frac{v_xv_y}{g}\right)(-v_y)-\left(-\frac{v_y^2}{2g}\right)(v_x)\right|$$

$$L=m\left|-\frac{v_xv_y^2}{g}+\frac{v_xv_y^2}{2g}\right|=\frac{mv_xv_y^2}{2g}$$

Substitute v_x and v_y:

$$L=\frac{m}{2g}\left(\frac{v_0}{2}\right)\left(\frac{\sqrt{3}v_0}{2}\right)^2=\frac{3mv_0^3}{16g}$$

(NCERT Ed. 2025-26, 11th, Page No. 106)

7. (2)

Equation of Continuity:

$$A_Av_A=A_Bv_B\Rightarrow 40v_A=10v_B\Rightarrow v_B=4v_A$$

Bernoulli Equation: $\Delta P=\frac{1}{2}\rho(v_B^2-v_A^2)$ (horizontal pipe)

$$7500=\frac{1}{2}(1000)[(4v_A)^2-v_A^2]$$

$$7500=500(15v_A^2)\Rightarrow 7500=7500v_A^2$$

$$\Rightarrow v_A=1.0\text{ m/s}$$

(NCERT Ed. 2025-26, 11th, Page No. 188)

[Contd...

8. (2)

Original efficiency $\eta_0 = 1 - \frac{T_2}{T_1}$.

New temperatures: $T_1' = 1.2T_1, T_2' = 0.8T_2$.

New efficiency $\eta' = 1 - \frac{0.8T_2}{1.2T_1} = 1 - \frac{2}{3}\frac{T_2}{T_1}$.

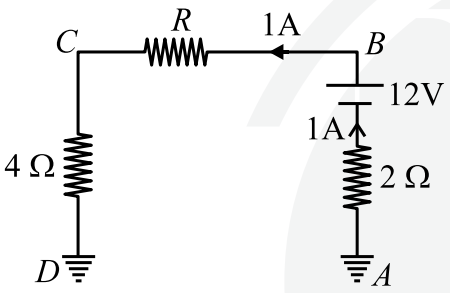
Given $\eta' = \frac{5}{3}\eta_0 \Rightarrow 1 - \frac{2}{3}(1-\eta_0) = \frac{5}{3}\eta_0$.

$1 - \frac{2}{3} + \frac{2}{3}\eta_0 = \frac{5}{3}\eta_0 \Rightarrow \frac{1}{3} = \eta_0$.

Efficiency $= \frac{1}{3} = 33.3\%$.

(NCERT Ed. 2025-26, 11th, Page No. 238)

9. (1)



From $A \to B \to C \to D$

$0 - (2\times1) + 12 - (R\times1) - (4\times1) = 0$

$\Rightarrow 6 - R = 0$

$\Rightarrow R = 6\,\Omega$

(NCERT Ed. 2025-26, 12th, Page No. 98)

10. (3)

Initial Case:

$E = hn - f$

From this, we can find the energy of the incident photon: $hn = E + f$

Second Case (Frequency is doubled, so $\nu = 2n$):

Let the new maximum kinetic energy be E'.

$E' = h(2n) - f$

$E' = 2(hn) - f$

Substitute the value of hn from the initial case:

$E' = 2(E + f) - f$

$E' = 2E + 2f - f$

$E' = 2E + f$

(NCERT Ed. 2025-26, 12th, Page No. 281)

11. (2)

Change in velocity (Δv) = Area under the *a-t* graph.

Total Area $= (2\times0) + (2\times3) + (2\times6)$

$= 0 + 6 + 12 = 18$ m/s

Final Velocity = Initial velocity

$+\Delta v = 2.00 + 18 = 20.0$ m/s

(NCERT Ed. 2025-26, 11th, Page No. 22)

12. (1)

Statement I is correct: The experiment proved the existence of a dense, positively charged nucleus.

Statement II is incorrect: Rutherford's model failed to explain atom stability (accelerating electrons should radiate energy and spiral into the nucleus) and could not explain discrete emission spectra (it predicted a continuous spectrum).

(NCERT Ed. 2025-26, 12th, Page No. 297)

13. (2)

We know the volume (V) is $0.050\ \text{m}^3$ and the height (h) is 0.500 m.

$A = \frac{V}{h} = \frac{0.050\ \text{m}^3}{0.500\ \text{m}} = 0.100\ \text{m}^2$

Since the cylinder is in a vacuum, there is no atmospheric pressure pushing down. The only downward force is the weight of the mass on top.

$F = m\times g = 10.0\ \text{kg}\times 9.8\ \text{m/s}^2 = 98\ \text{N}$

$P = \frac{F}{A} = \frac{98\ \text{N}}{0.100\ \text{m}^2} = 980\ \text{Pa}$

(NCERT Ed. 2025-26, 11th, Page No. 181)

14. (1)

End *C* is at distance *L* from pivot *P*. End *D* is at distance 2*L* from pivot *P*.

Using the formula for motional emf of a rotating rod

$V = \frac{1}{2}B\omega x^2 : V_P - V_C = \frac{1}{2}B\omega(L)^2 = 0.5B\omega L^2$.

$V_P - V_D = \frac{1}{2}B\omega(2L)^2 = 2B\omega L^2$.

Potential difference

$|V_D - V_C| = |(V_P - V_C) - (V_P - V_D)|$

$= |0.5B\omega L^2 - 2B\omega L^2| = 1.5B\omega L^2$

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[Contd...

15. (2)

Standard wave equation $E = E_0 \sin(\omega t - kx)$.

From the given equation, $\omega = 10^8$ rad/s and $k = 0.5$ rad/m.

Speed of the wave in the medium

$v = \frac{\omega}{k} = \frac{10^8}{0.5} = 2 \times 10^8$ m/s.

Refractive index $n = \frac{c}{v} = \frac{3 \times 10^8}{2 \times 10^8} = 1.5$.

(NCERT Ed. 2025-26, 12th, Page No. 212)

16. (2)

Cutting a lens parallel to the principal axis does not change its focal length. Therefore, f = +20 cm for the half-lens.

Sign Convention: Object distance u = –30 cm.

Lens Formula: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$\frac{1}{20} = \frac{1}{v} - \frac{1}{-30}$

$\frac{1}{v} = \frac{1}{20} - \frac{1}{30} = \frac{3-2}{60} = \frac{1}{60}$

Result: v = +60 cm (Positive sign indicates a real image).

(NCERT Ed. 2025-26, 12th, Page No. 235)

17. (3)

For an electron accelerated through potential V, its kinetic energy is eV.

The de Broglie wavelength $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m(eV)}}$.

Thus, $\lambda \propto \frac{1}{\sqrt{V}}$.

If V is increased by a factor of 4 (becomes 4V), then $\sqrt{V}$ increases by a factor of 2.

The new wavelength $\lambda' = \frac{\lambda}{2}$

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18. (2)

The orbiting electron constitutes a circular current

$I = \frac{e}{T} = \frac{e}{(2\pi r / v)} = \frac{ev}{2\pi r}$.

The magnetic field at the center of a circular current loop is $B = \frac{\mu_0 I}{2r}$.

Substituting I: $B = \frac{\mu_0 (ev / 2\pi r)}{2r} = \frac{\mu_0 ev}{4\pi r^2}$.

(NCERT Ed. 2025-26, 12th, Page No. 116)

19. (2)

Gauss's Law: Total electric flux.

The charge q is outside the small cube, so the net charge enclosed by the small cube is zero.

$\Phi = \frac{0}{\epsilon_0} = 0$

(NCERT Ed. 2025-26, 12th, Page No. 30)

20. (1)

From Coulomb's law, $F = \frac{1}{4\pi\epsilon_0} \times \frac{q_1 q_2}{r^2}$

The product $F\epsilon_0 = \frac{q^2}{4\pi r^2}$

The dimensional formula is

$\frac{[\text{Charge}]^2}{[\text{Distance}]^2} = \frac{[A^2T^2]}{[L^2]} = [M^0L^{-2}T^2A^2]$

(NCERT Ed. 2025-26, 11th, Page No. 10)

21. (1)

Magnitude of electric field intensity $\propto$ field line density

$\therefore E_A < E_B$

(NCERT Ed. 2025-26, 12th, Page No. 19)

22. (1)

Shortest path (zero drift): He must swim at an angle θ upstream such that

$2v \sin\theta = v \Rightarrow \sin\theta = \frac{1}{2} \Rightarrow \theta = 30°$

Velocity along width $= 2v\cos 30° = v\sqrt{3}$

Time $t_1 = \frac{W}{v\sqrt{3}}$

Shortest time: He swims directly across.

Velocity along width $= 2v$

Time $t_2 = \frac{W}{2v}$

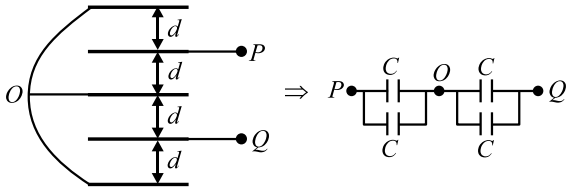
Ratio $\frac{t_1}{t_2} = \frac{W}{v\sqrt{3}} \times \frac{2v}{W} = \frac{2}{\sqrt{3}}$

(NCERT Ed. 2025-26, 11th, Page No. 46)

[Contd...

23. (3)

$C_{eq} = C = \frac{\epsilon_0 A}{d}$



(NCERT Ed. 2025-26, 12th, Page No. 72)

24. (1)

The process equation is $PV^{-2} = \text{constant}$.

This is a polytropic process $PV^x = \text{constant}$ with $x = -2$.

Molar heat capacity $C = C_v + \frac{R}{1-x}$.

For a diatomic gas, $C_v = \frac{5R}{2}$.

$C = \frac{5R}{2} + \frac{R}{1-(-2)} = \frac{5R}{2} + \frac{R}{3} = \frac{15R+2R}{6} = \frac{17R}{6}$.

(NCERT Ed. 2025-26, 11th, Page No. 231)

25. (3)

The induced emf is $\varepsilon = -\frac{d\phi}{dt}$, so current $i = \frac{\varepsilon}{R}$ vanishes when $\frac{d\phi}{dt} = 0$.

$\frac{d\phi}{dt} = t^2 - 6t + 9 = (t-3)^2$.

Thus $I = 0$ at $t = 3$ s

(NCERT Ed. 2025-26, 12th, Page No. 158)

26. (2)

On the axis of a dipole,

$B = \frac{\mu_0}{4\pi}\frac{2m}{r^3} \Rightarrow m = \frac{Br^3}{2(\mu_0/4\pi)}$

With r = 0.10m, $r^3 = 10^{-3}$ m^3

Hence $m = \frac{(2.0\times10^{-4})(10^{-3})}{2\times10^{-7}}$

$= \frac{2.0\times10^{-7}}{2\times10^{-7}} = 1.0$ Am^2

(NCERT Ed. 2025-26, 12th, Page No. 140)

27. (4)

$\mu \times \sin 60 = \mu_g \times \sin 30$

$\mu_g = \sqrt{3}$

$L_{AO} = \frac{a}{\cos\theta} = 2a, \quad L_{OB} = \frac{b}{\sin 30} = \frac{2b}{\sqrt{3}}$

$L_{op}t = (\mu_{voc} \times L_{AO}) \times (\mu_g \times L_{OB})$

$= (1\times 2a) + \left(\sqrt{3} \times \frac{2b}{\sqrt{3}}\right) = 2a + 2b$

(NCERT Ed. 2025-26, 12th, Page No. 228)

28. (2)

Springs are in series:

$\frac{1}{k_{eq}} = \frac{1}{3k} + \frac{1}{6k} = \frac{2+1}{6k} = \frac{1}{2k} \Rightarrow k_{eq} = 2k$

Time period $T = 2\pi\sqrt{\frac{m}{k_{eq}}} = 2\pi\sqrt{\frac{m}{2k}}$

(NCERT Ed. 2025-26, 11th, Page No. 272)

29. (2)

(1) Fundamental of a string: The standard formula is $f = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$. (A $\rightarrow$ III)

(2) Fundamental of a closed pipe: The wavelength is 4L, so the frequency is $f = \frac{v}{4L}$. (B $\rightarrow$ II)

(3) First overtone of an open pipe: This is the second harmonic. $f_2 = 2\times f_1 = 2\times\left(\frac{v}{2L}\right) = \frac{v}{L}$.

(C $\rightarrow$ IV)

(4) First overtone of a closed pipe: This is the third harmonic. $f_3 = 3\times f_1 = 3\times\left(\frac{v}{4L}\right) = \frac{3v}{4L}$. (D $\rightarrow$ I)

(NCERT Ed. 2025-26, 11th, Page No. 293)

30. (4)

Aluminium is a trivalent impurity (3 valence electrons). When added to silicon (4 valence electrons), it causes the following:

Option 2 is true: It creates a p-type semiconductor.

[Contd...

Option 3 is true: It introduces "holes" as majority charge carriers, so holes outnumber free electrons.
Option 1 is true: The addition of these extra charge carriers increases conductivity, which decreases resistance.

(NCERT Ed. 2025-26, 12th, Page No. 331)

31. (1)

$$E = \frac{V}{d}$$

$$E = \frac{0.50}{5.0 \times 10^{-7}}$$

$$E = 0.1 \times 10^{7}$$

$$E = 1.0 \times 10^{6} \text{ V/m}$$

(NCERT Ed. 2025-26, 12th, Page No. 334)

32. (2)

(1) Heat Gained (Cold Side):
Ice melting: $21 \times 80 = 1680$ cal
Heating water + calorimeter (mass = 21 + 9 = 30 g)
$30 \times 1 \times 10 = 300$ cal
Total Gained $= 1680 + 300 = 1980$ cal
(2) Heat Lost (Steam Side):
Condensation + cooling:
$m \times 540 + m \times 1 \times (100 - 10) = 630m$
(3) Equating:

$$630m = 1980 \Rightarrow m = \frac{198}{63} \approx 3.14 \text{ g}$$

(NCERT Ed. 2025-26, 11th, Page No. 213)

33. (2)

$$I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$$

Substitute our values:

$$I_{max} = (\sqrt{9I_0} + \sqrt{1I_0})^2 = (3\sqrt{I_0} + 1\sqrt{I_0})^2$$

$$= (4\sqrt{I_0})^2 = 16I_0$$

$$I_{min} = (\sqrt{9I_0} - \sqrt{1I_0})^2 = (3\sqrt{I_0} - 1\sqrt{I_0})^2$$

$$= (2\sqrt{I_0})^2 = 4I_0$$

Evaluate the target ratio:

$$\text{Ratio} = \frac{I_{max} - I_{min}}{I_{max} + I_{min}}$$

$$\text{Ratio} = \frac{16I_0 - 4I_0}{16I_0 + 4I_0}$$

$$\text{Ratio} = \frac{12I_0}{20I_0} = \frac{12}{20} = \frac{3}{5}$$

(NCERT Ed. 2025-26, 12th, Page No. 264)

34. (3)

Nuclear mass is proportional to A, and volume is proportional to R^3 . Since $R = R_0 A^{1/3}$ (the Reason), volume is also proportional to *A*. Therefore, $\text{Density} = \frac{\text{Mass}}{\text{Volume}}$ becomes independent of *A*, proving the Assertion.

(NCERT Ed. 2025-26, 12th, Page No. 309)

35. (2)

$$m_c v_c R = I_r \omega_r$$

$$(0.18 \text{ kg})(0.80 \text{ m s}^{-1})(0.50 \text{ m}) = (0.20 \text{ kg m}^2) \cdot \omega_r$$

$$0.072 \text{ kg m}^2 \text{ s}^{-1} = 0.20 \cdot \omega_r$$

$$\omega_r = \frac{0.072}{0.20} = 0.36 \text{ rad s}^{-1}$$

(NCERT Ed. 2025-26, 11th, Page No. 123)

36. (2)

$$N = mg\cos(\theta)$$

$$m = \frac{N}{g\cos(\theta)}$$

$$m = \frac{15}{10 \times \cos(60^\circ)}$$

m = 3 kg

(NCERT Ed. 2025-26, 11th, Page No. 61)

37. (2)

For ductile material- This means there is a large gap (elongation) between the yield point and the point where it actually snaps (Fracture Point).
For brittle material - break shortly after the elastic limit with very little plastic deformation. The Fracture point is very close to yield point.

(NCERT Ed. 2025-26, 11th, Page No. 169)

[Contd...

38. (3)

$\tau = F \times R = 12 \times 0.5 = 6\text{N} \cdot \text{m}$

$I = \frac{1}{2}mR^2 = \frac{1}{2} \times 2 \times (0.5)^2 = 0.25\text{kg} \cdot \text{m}^2$

$\alpha = \frac{\tau}{I} = \frac{6}{0.25} = 24\text{rad/s}^2$

$\theta = \frac{1}{2}\alpha t^2 = \frac{1}{2} \times 24 \times (4)^2 = 192\text{rad}$

$n = \frac{\theta}{2\pi} = \frac{192}{6} = 32$

(NCERT Ed. 2025-26, 11th, Page No. 120)

39. (2)

$E = -\frac{13.6}{n^2}z^2$

$-122.4 = -13.6z^2$

$z = 3$

(NCERT Ed. 2025-26, 12th, Page No. 303)

40. (3)

For 10 V D.C.

$V = IR \Rightarrow R = \frac{10}{2.5} = 4\,\Omega$

For 10 V A.C.

$V = IZ \Rightarrow Z = \frac{V}{I} = \frac{10}{2} = 5\,\Omega$

$\therefore Z = \sqrt{R^2 + X_L^2} = 5$

$\Rightarrow R^2 + X_L^2 = 25$

$\Rightarrow X_L^2 = 5^2 - 4^2$

$\Rightarrow X_L = 3\,\Omega$

(NCERT Ed. 2025-26, 12th, Page No. 190)

41. (2)

Mass $M = \frac{4}{3}\pi R^3 \rho \Rightarrow R \propto \left(\frac{M}{\rho}\right)^{1/3}$

For the planet, M is $8x$ and ρ is $8x$, so

$\frac{R_p}{R_e} = \left(\frac{8}{8}\right)^{1/3} = 1$. Both have the same radius.

Escape velocity $v_e = \sqrt{\frac{2GM}{R}} \propto \sqrt{M}$.

Since M is 8 times larger and R is the same,

$\frac{v_{ep}}{v_{ee}} = \sqrt{8} = 2\sqrt{2}$

(NCERT Ed. 2025-26, 11th, Page No. 140)

42. (2)

$y_1 = 2\sin 500\pi t, y_2 = 1\sin 506\pi t$

$500\pi = 2\pi\nu_1 \Rightarrow \nu_1 = 250$ Hz

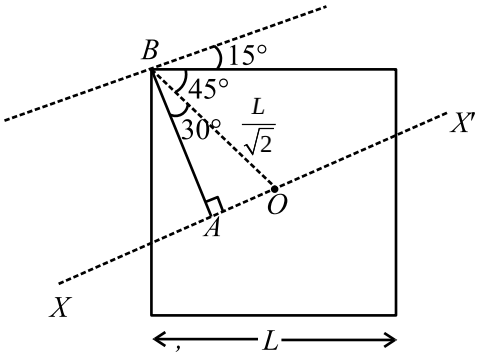
$506\pi = 2\pi\nu_2 \Rightarrow \nu_2 = 253$ Hz

$\nu = \nu_2 - \nu_1 = 253 - 250 = 3$ beats/s

Number of beats per minute $= 3 \times 60 = 180$

(NCERT Ed. 2025-26, 11th, Page No. 296)

43. (2)



Draw imaginary line XX' using through centre O and parallel to actual axis.

In ΔABO, $\cos 30° = \frac{AB}{\frac{L}{\sqrt{2}}}$

$AB = \frac{L}{\sqrt{2}} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2\sqrt{2}}L$

$I_{XX'} = \frac{ML^2}{6} - \frac{ML^2}{12}$

Using parallel axis theorem,

$I = I'_{XX} + M(AB^2)$

$= \frac{ML^2}{12} + M\left(\frac{\sqrt{3}L}{2\sqrt{2}}\right)^2$

$= \frac{11ML^2}{24}$

(NCERT Ed. 2025-26, 11th, Page No. 87)

44. (3)

According to Maxwell's law of distribution of speeds, Speed of molecules of gas can be any value between zero and infinity and most probable speed increases with increase in temperature of gas.

(NCERT Ed. 2025-26, 11th, Page No. 256)

45. (1)

As, $\Delta W_{AB} = p\Delta V = 10(2-1) = 10\text{J}$

and $\Delta W_{BC} = 0$, because V is constant

From first law of thermodynamics,

$\Delta Q = \Delta W + \Delta U$

As $ABCA$ is a cyclic process, therefore

$\Delta U = 0$

$\therefore \Delta Q = \Delta W_{AB} + \Delta W_{BC} + \Delta W_{CA} = \Delta W_{AB} + \Delta W_{CA}$

or $\Delta W_{CA} = \Delta Q - \Delta W_{AB} = 5 - 10 = -5\text{J}$

(NCERT Ed. 2025-26, 11th, Page No. 235)

46. (4)

$$\text{Av. } C_{v,m} = \frac{n_1 C_{v,m_1} + n_2 C_{v,m_2}}{n_1 + n_2}$$

For adiabatic process dU = dW

$$\frac{dT}{T} = -\frac{R}{C_{v,m}}\left(\frac{dV}{V}\right)$$

$$n_1 C_{v,m_1} dT + n_2 C_{v,m_2}\, dT = -(n_1 RT + n_2 RT) \times \frac{dV}{V}$$

$$\ln \frac{T_2}{T_1} = -\frac{1}{2}\ln\left(\frac{V_2}{V_1}\right) \Rightarrow T_2 = 320 \times \left(\frac{1}{4}\right)^{1/2} = 160\text{ K}$$

$$\Delta U = (n_1 C_{v,m_1} + n_2 C_{v,m_2})\Delta T = -960\text{ R}$$

(NCERT Ed. 2025-26, 11th, Page No. 144)

47. (3)

Diamond is covalent, yet it has high melting point because diamond has a three-dimensional network involving strong C–C bonds, which are very difficult to break.

(NCERT Ed. 2022-23, 11th, Page No. 326)

48. (3)

For strong electrolytes, conductivity decreases with decrease in concentration because on dilution total number of ions per unit volume of solution decreases.

(NCERT Ed. 2025-26, 12th, Page No. 46)

49. (4)

Energy of photon corresponding to second line of Balmer series for Li^{2+} ion

$$= (13.6) \times (3)^2 \left[\frac{1}{2^2} - \frac{1}{4^2}\right] = 13.6 \times \frac{27}{16}$$

Energy needed to eject electron from n=2 level in H-atom;

$$= 13.6 \times 1^2 \times \left[\frac{1}{2^2} - \frac{1}{\infty^2}\right] \Rightarrow \frac{13.6}{4}$$

K.E. of ejected electron

$$= 13.6 \times \frac{9 \times 3}{16} - \frac{13.6}{4} = 13.6 \times \left(\frac{27-4}{16}\right) \Rightarrow 19.55\text{ eV}$$

(NCERT Ed. 2025-26, 11th, Page No. 45)

50. (1)

$$\Delta T_f = i_1 k_f m_1 + i_2 k_f m_2$$

$$= 1 \times 1.86 \times \frac{0.5}{0.5} + 1 \times 1.86 \times \frac{0.5}{0.5} = 3.72°\text{C}$$

$$T_f = -3.72°\text{C}$$

(NCERT Ed. 2025-26, 12th, Page No. 18)

51. (2)

Wilkinson catalyst, $[(Ph_3P)_3RhCl]$ is used as catalyst for hydrogenation of alkenes.

(NCERT Ed. 2025-26, 12th, Page No. 137)

52. (2)

The presence of $–NO_2$ group at ortho- and para-positions increases the reactivity of haloarenes towards ArS_N reactions. It is because $–NO_2$ group withdraws the electron density from the benzene ring thus facilitates the attack of the nucleophile on haloarenes also the carbanion formed is stabilized through resonance.

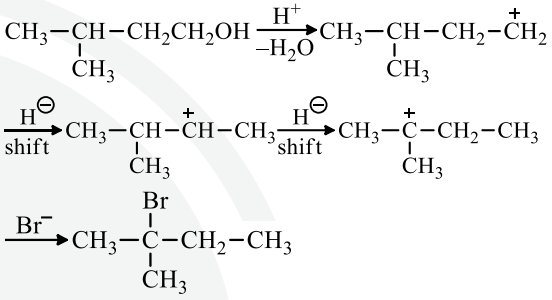
(NCERT Ed. 2025-26, 12th, Page No. 183)

53. (1)

1s orbital is closest to the nucleus, so it would not be allow to emit a photon.

(NCERT Ed. 2025-26, 11th, Page No. 47)

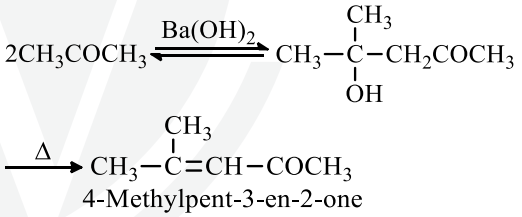
54. (4)



(NCERT Ed. 2025-26, 12th, Page No. 224)

55. (4)

Aldol condensation:



4-Methylpent-3-en-2-one

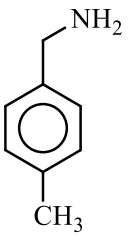
(NCERT Ed. 2025-26, 12th, Page No. 241)

56. (1)

- A system at higher temperature has more randomness in it than one at lower temperature.
- When a system is in equilibrium, the entropy is maximum.

(NCERT Ed. 2025-26, 11th, Page No. 159)

57. (2)

Lone pair of nitrogen in  is localised, so it is most basic in nature.

(NCERT Ed. 2025-26, 12th, Page No. 269)

[Contd...

58. (1)

$Zn(OH)_2 \rightleftharpoons \underset{S}{Zn^{2+}} + \underset{2S}{2OH^-}$

$K_{sp} = (S)(2S)^2 = 4S^3$

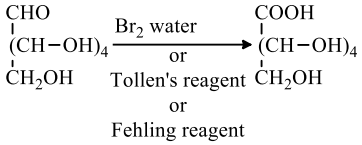
For pH = 12, pOH = 14 –12 = 2

$\therefore [OH^-] = 10^{-2}\,M \quad \therefore K_{sp} = (S')(10^{-2})^2$

$4S^3 = S' \times 10^{-4}; \quad S' = 4 \times 10^4 S^3 \text{ mol } L^{-1}$

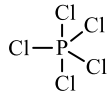
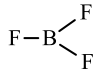
(NCERT Ed. 2025-26, 11th, Page No. 204)

59. (1)



(NCERT Ed. 2025-26, 12th, Page No. 283)

60. (2)

- $:\ddot{\underset{\cdot\cdot}{Cl}}-S-\ddot{\underset{\cdot\cdot}{Cl}}:$ → Complete octet
-  → 10 electrons around P-atom (expanded octet)
- $\dot{N}=O$ → Odd electron molecule
-  → sp^2 (trigonal planar)

(NCERT Ed. 2025-26, 11th, Page No. 105)

61. (4)

Among 3d series, Zn has lowest enthalpy of atomisation.

(NCERT Ed. 2025-26, 12th, Page No. 94)

62. (3)

	2X	+ Y	⟶ X_2Y
(*i*)	0.1 M	0.2 M	–
(f)	0.1 – 0.02 = 0.08 M	0.2 – 0.01 = 0.19 M	

$\text{rate} = k[X]^2[Y]$

$= 1 \times 10^{-4} \times (0.08)^2 (0.19) = 1.216 \times 10^{-7}\ Ms^{-1}$

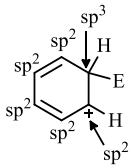
(NCERT Ed. 2025-26, 12th, Page No. 67)

63. (1)

For strong acid (HCl) and strong base (NaOH), heat of neutralisation is maximum.

(NCERT Ed. 2025-26, 11th, Page No. 156)

64. (2)



(NCERT Ed. 2025-26, 11th, Page No. 323)

65. (3)

$\underset{\substack{n_f=5\\16\text{ mL}\\0.05\text{ M}}}{KMnO_4} + \underset{\substack{n_f=2\\40\text{ mL}\\XM}}{C_2O_4^{2-}} \xrightarrow{H^+} Mn^{2+} + CO_2$

M,eq. of $C_2O_4^{2-}$ = m. eq. of $KMnO_4$

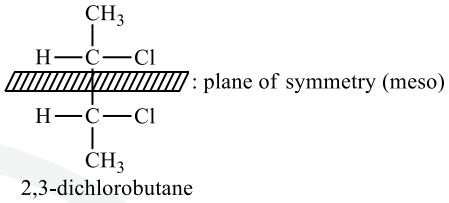
2(40)X = 5 × 16 × 0.05

X = 0.05 M $\quad \therefore [H^+] = 2X = 0.1\,M$

pH = 1

(NCERT Ed. 2025-26, 11th, Page No. 249)

66. (4)



2,3-dichlorobutane

(NCERT Ed. 2025-26, 12th, Page No. 179)

67. (2)

Kjeldahl method is not applicable to compounds containing nitrogen in nitro and azo groups and nitrogen present in ring.

(NCERT Ed. 2025-26, 11th, Page No. 288)

68. (4)

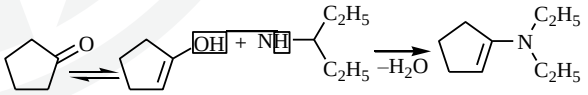
$H_2SO_4 \rightarrow 2H^+ + SO_4^{2-}$

$\text{Cathode}: 2H^+ + 2e^- \rightarrow H_2$

$\text{Anode}: \ 2H_2O \rightarrow O_2 + 4H^+ + 4e^-$

(NCERT Ed. 2025-26, 12th, Page No. 53)

69. (1)



(NCERT Ed. 2025-26, 12th, Page No. 238)

70. (1)

$$r = \frac{-nRT\ln\frac{12}{24}}{-P_{ext}nRT\left(\frac{1}{P_2}-\frac{1}{P_1}\right)}$$

$$r = \frac{-0.7}{24\left(\frac{1}{24}-\frac{1}{12}\right)} = \frac{0.7\times 24}{24(1)}$$

10r = 10 × 0.7 = 7

(NCERT Ed. 2025-26, 11th, Page No. 141)

71. (3)

More negative $\Delta_f H°$, represents more stable compound.

(NCERT Ed. 2025-26, 11th, Page No. 149)

[Contd...

72. (4)

$CH_3-\bar{C}H_2$ is least stable due to +I effect of methyl group.

(NCERT Ed. 2025-26, 11th, Page No. 272)

73. (1)

$$CH_3-MgCl \xrightarrow{CO_2} CH_3-COOMgCl \xrightarrow{H_3O^+}$$
$$CH_3COOH \xrightarrow{LiAlH_4} CH_3-CH_2OH$$

(NCERT Ed. 2025-26, 12th, Page No. 253)

74. (3)

$C_1=C_2$

$$\frac{\left(\frac{2}{M}\right)}{100\times10^{-3}} = \frac{\left(\frac{3}{60}\right)}{100\times10^{-3}} \quad \therefore M = \frac{2\times60}{3} = 40\ g\ mol^{-1}$$

(NCERT Ed. 2025-26, 12th, Page No. 22)

75. (3)

Boiling point of n-pentane is more than that of isobutane because isobutane is more spherical than n-pentane as a result there are less interaction forces between isobutane molecules.

(NCERT Ed. 2025-26, 11th, Page No. 302)

76. (2)

More the stability of intermediate carbocation, more will be reactivity of alkene towards hydration.

(NCERT Ed. 2025-26, 12th, Page No. 199)

77. (3)

$B_2 : (\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2\ (\pi 2p_x^1 \equiv \pi 2p_y^1)$

Due to presence of two unpaired electron in π2p orbitals, B_2 is paramagnetic in nature.

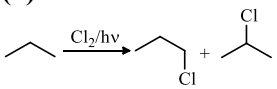
(NCERT Ed. 2025-26, 11th, Page No. 131)

78. (3)

SI unit of density is Kg m^{-3}.

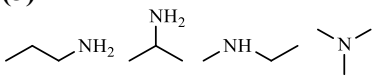
(NCERT Ed. 2025-26, 11th, Page No. 10)

79. (3)



(NCERT Ed. 2025-26, 11th, Page No. 303)

80. (3)



(NCERT Ed. 2025-26, 12th, Page No. 261)

81. (2)

All the actinoids elements are radioactive.

(NCERT Ed. 2025-26, 12th, Page No. 100)

82. (3)

$$\ln\frac{[R]}{[R_0]} = -kt; \qquad 2.303\log\frac{[R]}{[R_0]} = -kt$$

$$\log\frac{[R]}{[R_0]} = -\frac{kt}{2.303}$$

(NCERT Ed. 2025-26, 12th, Page No. 74)

83. (2)

3 moles of Grignard reagent required.

1 mole for SH, 1 mole for aldehyde and 1 mole for ketone

(NCERT Ed. 2025-26, 12th, Page No. 236)

84. (3)

40% aq. solution of HCHO is known as formalin.

(NCERT Ed. 2025-26, 12th, Page No. 243)

85. (1)

	$CO_2(g)$ +	$C(s) \rightleftharpoons 2CO(g)$
(*i*)	0.4 atm	–
(f)	(0.4 – x) atm	2x atm

$p_{total} = 0.5 = 0.4 - x + 2x$

$0.5 = 0.4 + x; \qquad x = 0.1\ atm$

$$\therefore K_p = \frac{(2x)^2}{(0.4-x)} = \frac{0.2\times0.2}{0.3} = 0.13\ atm$$

(NCERT Ed. 2025-26, 11th, Page No. 184)

86. (3)

Number of chiral carbon is four.

(NCERT Ed. 2025-26, 12th, Page No. 286)

87. (1)

HVZ reaction is used for the preparation of α-halo acids [X=Cl, Br]

(NCERT Ed. 2025-26, 12th, Page No. 253)

88. (4)

- $S_2F_2 : 2x - 2 = 0 \Rightarrow x = +1$
- $N_2H_4 : 2x + 4 = 0 \Rightarrow x = -2$
- $H_3PO_3 : 3 + x - 6 = 0 \Rightarrow x = +3$
- $IF_5 : x - 5 = 0 \Rightarrow x = +5$

(NCERT Ed. 2025-26, 11th, Page No. 240)

89. (1)

3° alkyl halide undergo elimination reaction with strong base on heating.

(NCERT Ed. 2025-26, 12th, Page No. 216)

90. (1)

Let wt. of NH_4NO_3 and $(NH_4)_2HPO_4$ are x and y gram respectively

$$\frac{\frac{x}{80}\times2\times14 + \frac{y}{132}\times2\times14}{x+y}\times100 = 30.4 \Rightarrow x:y = 2:1$$

(NCERT Ed. 2025-26, 11th, Page No. 18)

[Contd...

91. (3)

The physico-chemical approach to study and understand living organisms is called "Reductionist Biology".

(NCERT Ed. 2025-26, 11th, Page No. 85)

92. (2)

In addition to the genomic DNA (the single chromosome/circular DNA), many bacteria have small circular DNA outside the genomic DNA. These smaller DNA are called plasmids. The cell envelope consists of a tightly bound three layered structure i.e., the outermost glycocalyx followed by the cell wall and then the plasma membrane.

(NCERT Ed. 2025-26, 11th, Page No. 90)

93. (4)

RNA is the genetic material in QB bacteriophage.

(NCERT Ed. 2025-26, 12th, Page No. 87)

94. (1)

Pyrenoids are storage bodies containing protein and starch found in the chloroplasts of most green algae, such as *Volvox*. The prothallus is the inconspicuous, small, multicellular, free-living, mostly photosynthetic thalloid gametophyte stage of pteridophytes like *Equisetum*. The life cycle of a moss (like *Polytrichum*) involves a juvenile stage called the protonema, which develops directly from a spore. Gemmae are green, multicellular, asexual buds which develop in small receptacles called gemma cups on the thalli of liverworts like *Marchantia*.

(NCERT Ed. 2025-26, 11th, Page No. 29)

95. (3)

Pyramid of energy is always upright, can never be inverted, because when energy flows from a particular trophic level to the next trophic level, some energy is always lost as heat at each step.

(NCERT Ed. 2025-26, 12th, Page No. 213)

96. (2)

In a monohybrid cross, from the Punnett square it is easily seen that 1/4th of the random fertilisations lead to TT, 1/2 lead to Tt and 1/4th to tt. This leads to a phenotypic ratio of 3/4th tall : (1/4 TT + 1/2 Tt) and 1/4th tt, i.e., a 3:1 ratio, but a genotypic ratio of 1:2:1.

The ratio of heterozygous tall plants (Tt) is 1/2 or 50%. So, 1/2 of 1200 = 600.

(NCERT Ed. 2025-26, 12th, Page No. 58)

97. (2)

The innermost layer of the cortex is called endodermis. It comprises a single layer of barrel-shaped cells without any intercellular spaces. The tangential as well as radial walls of the endodermal cells have a deposition of water-impermeable, waxy material suberin in the form of casparian strips.

(NCERT Ed. 2025-26, 11th, Page No. 74)

98. (1)

The central cell after triple fusion becomes the primary endosperm cell (PEC) and develops into the endosperm while the zygote develops into an embryo.

(NCERT Ed. 2025-26, 12th, Page No. 18)

99. (3)

Streptokinase produced by the bacterium *Streptococcus* and modified by genetic engineering is used as a 'clot buster' for removing clots from the blood vessels of patients who have undergone myocardial infarction leading to heart attack.

(NCERT Ed. 2025-26, 12th, Page No. 153)

100. (4)

Species: *tuberosum* (A)
Genus: *Solanum* (E)
Family: Solanaceae (D)
Order: Polymoniales (C)
Kingdom: Plantae (B)

(NCERT Ed. 2025-26, 11th, Page No. 6)

101. (1)

Mutualism: Both species benefit (+, +)
Predation: One species benefits, the other is harmed (+, –)
Commensalism: One species is benefitted and the other is neither benefitted nor harmed (+, 0)
Amensalism: One species is harmed whereas the other is unaffected (–, 0)

(NCERT Ed. 2025-26, 12th, Page No. 197)

[Contd...

102. (2)

During the conversion of Fructose-6-phosphate into Fructose 1, 6-bisphosphate, ATP is utilised. Water is formed during the conversion of phosphoenolpyruvate from 2-phosphoglycerate.

(NCERT Ed. 2025-26, 11th, Page No. 156)

103. (2)

Both C_3 and C_4 plants use the Calvin cycle to synthesize sugar (glucose). In C plants, the initial CO_2 fixation occurs in the mesophyll cells, but the final fixation and sugar synthesis via the Calvin cycle take place in the bundle sheath cells. In C_3 plants, the primary CO_2 acceptor is RuBP (a 5-carbon compound). In C_4 plants, the primary acceptor is PEP (a 3-carbon compound). Kranz anatomy is a specialized leaf structure found only in C_4 plants, where bundle sheath cells are arranged in a wreath-like manner. Generally, C_4 plants are more efficient at photosynthesis in high light and temperature conditions and typically produce more biomass compared to C_3 plants, which lose energy through a process called photorespiration.

(NCERT Ed. 2025-26, 11th, Page No. 147)

104. (3)

Honey bees have a haplodiploid sex-determination system. While female honey bees (queens and workers) are diploid (2n = 32), drone honey bees (males) develop from unfertilized eggs and are haploid (n = 16). Therefore, every somatic cell in a drone starts with 16 chromosomes.

The G_1 phase (Gap 1) is the period before DNA replication. The cell contains the basic haploid set of chromosomes. Number of chromosomes = 16.

During the S phase (which occurs between G_1 and G_2), DNA replication takes place. While the amount of DNA doubles (the chromatids double), the number of chromosomes does not change. In G_2, each of the 16 chromosomes consists of two sister chromatids, but they are still joined at a single centromere. Number of chromosomes = 16.

In Metaphase of mitosis, the chromosomes align at the equatorial plate. They are still composed of two sister chromatids attached at the centromere. The division of the centromere (which turns sister chromatids into individual chromosomes) does not occur until Anaphase. Number of chromosomes = 16.

(NCERT Ed. 2025-26, 11th, Page No. 121, 123)

105. (4)

Diatoms are the chief 'producers' in the oceans.

Dinoflagellates: Very often, red dinoflagellates (Example: *Gonyaulax*) undergo such rapid multiplication that they make the sea appear red (red tides).

Euglenoids: Instead of a cell wall, they have a protein rich layer called pellicle which makes their body flexible.

Slime moulds: Slime moulds are saprophytic protists.

(NCERT Ed. 2025-26, 11th, Page No. 14)

106. (1)

Ethephon promotes female flowers in cucumbers thereby increasing the yield.

(NCERT Ed. 2025-26, 11th, Page No. 177)

107. (4)

Colour blindness occurs in about 8 per cent of males and only about 0.4 per cent of females. Thalassemia differs from sickle-cell anaemia in that the former is a quantitative problem of synthesising too few globin molecules.

(NCERT Ed. 2025-26, 12th, Page No. 74)

108. (1)

In a typical embryo sac, three cells are grouped together at the micropylar end and constitute the egg apparatus. The egg apparatus, in turn, consists of two synergids and one egg cell. Three cells are at the chalazal end and are called the antipodals. These all are haploid cells and the antipodals degenerate after double fertilisation.

(NCERT Ed. 2025-26, 12th, Page No. 11)

109. (4)

Detritivores (e.g., earthworm) break down detritus into smaller particles. This process is called fragmentation.

(NCERT Ed. 2025-26, 12th, Page No. 207)

110. (4)

Some examples of recent extinctions include the dodo (Mauritius), quagga (Africa), thylacine (Australia), Steller's Sea Cow (Russia) and three subspecies (Bali, Javan, Caspian) of tiger.

(NCERT Ed. 2025-26, 12th, Page No. 221)

111. (2)

DNA polymerase uses a DNA template to catalyse the polymerisation of deoxynucleotides. The discontinuously synthesised DNA fragments are later joined by the enzyme DNA ligase. RNA polymerase catalyses transcription. Dnase digests DNA.

(NCERT Ed. 2025-26, 12th, Page No. 90)

112. (3)

Banyan tree has prop roots for support while maize has stilt roots for support.

(NCERT Ed. 2025-26, 11th, Page No. 67)

113. (1)

According to the chemiosmotic hypothesis in chloroplasts, the primary electron acceptor (of Photosystem II) is located toward the outer side (stroma side) of the thylakoid membrane. It transfers its electron not directly to an electron carrier, but to an H carrier (such as Plastoquinone). As this carrier transfers the electron, it also picks up a proton (H^+) from the stroma to move it into the lumen, contributing to the proton gradient. In cyclic photophosphorylation, only Photosystem I (PS I) is functional. The electrons are circulated within the photosystem and the electron transport chain, resulting only in the synthesis of ATP. Since there is no photolysis of water and electrons do not pass to $NADPH^+$, $NADPH + H^+$, is not synthesized.

(NCERT Ed. 2025-26, 11th, Page No. 141)

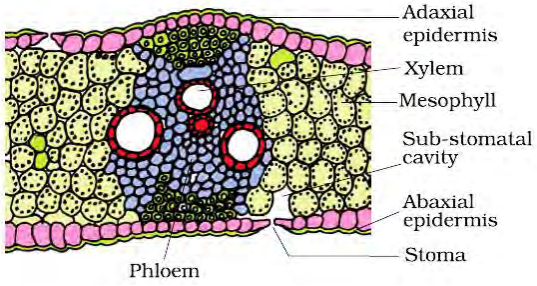
114. (4)

The full potential of penicillin as an effective antibiotic was established by Ernest Chain and Howard Florey. Ramdeo Misra is revered as Father of Ecology in India.

(NCERT Ed. 2025-26, 12th, Page No. 152)

115. (2)

The given diagram represents the T.S. of a monocot leaf. A (adaxial epidermis) and E (abaxial epidermis) both contain stomata and are covered by cuticle. B (xylem) and G (phloem) constitute the vascular tissue system. C (mesophyll) constitutes the ground (fundamental) tissue system. D (stoma) and F (sub-stomatal cavity) are a part of epidermal tissue system.



T.S. of leaf : Monocot

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116. (1)

Majority of plants use biotic agents for pollination. Pollination by water is quite rare in flowering plants and is limited to about 30 genera, mostly monocotyledons.

(NCERT Ed. 2025-26, 12th, Page No. 12)

117. (1)

In incomplete dominance, the F_1 generation has a phenotype that is in-between the two parents. In the inheritance of flower colour in Snapdragon, the F_2 phenotypic ratio from a cross between red (RR) and white (rr) flowered plants is 1 (Red) : 2 (Pink) : 1 (White).

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118. (3)

Acetyl CoA enters the aerobic respiratory pathway as a substrate for 1st reaction of TCA cycle. For each acetyl CoA entering the TCA pathway - 3 $NADH + H^+$, 1 $FADH_2$ and 1 GTP (that is coupled to conversion of ADP into ATP) are produced. This synthesis of ATP is substrate level phosphorylation. $NADH + H^+$ and $FADH_2$ produce ATP by undergoing oxidative phosphorylation in the ETS. Thus, for 1 molecule of acetyl CoA - 3 $NADH + H^+$, 1 $FADH_2$ produce 11 ($3\times3 + 1\times2$) ATP in the ETS. For 8 such molecules of acetyl CoA, a total of 88 ATP will be synthesised via oxidative phosphorylation.

(NCERT Ed. 2025-26, 11th, Page No. 159, 160)

119. (2)

According to the fluid mosaic model of the plasma membrane, the quasi-fluid nature of lipid enables lateral movement of proteins within the overall bilayer.

(NCERT Ed. 2025-26, 11th, Page No. 94)

120. (4)

The Mycoplasma are organisms that completely lack a cell wall. They are the smallest living cells known and can survive without oxygen. Many mycoplasma are pathogenic in animals and plants.

(NCERT Ed. 2025-26, 11th, Page No. 14)

121. (4)

In bryophytes, the sporophyte is not free-living but attached to the photosynthetic gametophyte and derives nourishment from it.

(NCERT Ed. 2025-26, 11th, Page No. 29)

122. (4)

Lactobacillus is used for the production of curd and increases vitamin B_{12} content in it. Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol lowering agents.

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[Contd...

123. (3)

Without the repressor protein, there is nothing to bind to the operator. Therefore, the RNA polymerase is free to bind to the promoter and transcribe the structural genes (*z*, *y* and *a*) continuously, regardless of whether lactose (the inducer) is present or absent.

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124. (3)

There are four major causes of biodiversity loss ('The Evil Quartet' is the sobriquet used to describe them). (i) Habitat loss and fragmentation, (ii) Over-exploitation, (iii) Alien species invasions, (iv) Co-extinctions.

(NCERT Ed. 2025-26, 12th, Page No. 222)

125. (4)

A large number of rounded, parenchymatous cells with large intercellular spaces which occupy the central portion of the stem constitute the pith in dicot stem.

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126. (3)

The stage between the two meiotic divisions is called interkinesis and is generally short lived. There is no replication of DNA during interkinesis.

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127. (2)

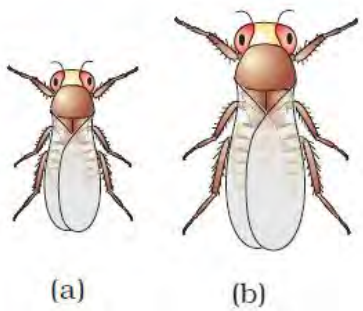
China rose is a member of family Malvaceae.

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128. (1)

The given fruitflies have XX and XY sex chromosomes, respectively. Under normal conditions, if they are crossed, the progeny will have either XX or XY sex chromosomes i.e. each progeny will have two sex chromosomes.

	X	Y
X	XX	XY
X	XX	XY



Drosophila melanogaster
(a) Male (b) Female

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129. (3)

Given data:
Initial Population = 375
Births (B) = 72
Immigration (I) = 32
Deaths (D) = 40
Emigration (E) = 56
Total number of individuals added to population = 72 + 32 = 104
Total number of individuals removed from population = 56 + 40 = 96
Net change in population = 104-96 = increase in 8 individuals.

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130. (4)

In most situations, ABA acts as an antagonist to GAs.

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131. (3)

The red algae usually reproduce vegetatively by fragmentation. They reproduce asexually by non-motile spores and sexually by non-motile gametes.

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132. (4)

Goat: Primary consumer (T_2)
Deer: Primary consumer (T_2)
Algae: Producer (T_1)
Man: As an omnivore that can eat secondary consumers (like certain birds or fish) or act as a top predator in many food chains, Man occupies the higher trophic levels, including the fourth trophic level.

(NCERT Ed. 2025-26, 12th, Page No. 211)

133. (2)

Bacteria are the most abundant micro-organisms.

(NCERT Ed. 2025-26, 11th, Page No. 12)

134. (3)

The process of copying genetic information from one strand of the DNA into RNA is termed as transcription.

(NCERT Ed. 2025-26, 12th, Page No. 91)

135. (3)

Axillary buds of stems may also get modified into woody, straight and pointed thorns. Thorns are found in many plants such as *Citrus*, *Bougainvillea*.

(NCERT Ed. 2025-26, 11th, Page No. 68)

136. (1)

Continuous culture keeps cells in the log (exponential) phase, where they are most active.

Draining the used medium and adding fresh medium maintains the optimum growth conditions, allowing the cells to remain in the active log/exponential phase, leading to a larger biomass and higher yields.

(NCERT Ed. 2025-26, 12th, Page No. 173)

137. (1)

Rh incompatibility usually does not affect the first pregnancy because the mother is not yet sensitized. Antibodies form after exposure during delivery.

Injection of anti-Rh antibodies (anti-D) after first delivery prevents formation of maternal antibodies, thus protecting future pregnancies.

Blood group O has both anti-A and anti-B antibodies in plasma. That's why it is a universal donor (RBCs lack antigens), not due to absence of antibodies.

During clotting, thrombin converts soluble fibrinogen into insoluble fibrin, forming the clot mesh.

The Sino-Atrial Node (SAN) acts as the pacemaker and generates 70–75 impulses per minute under normal conditions.

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138. (1)

The frequency of heterozygotes (2pq) = 640/2000 = 0.32. Therefore, pq = 0.16. The values for p and q that satisfy p + q = 1 and pq = 0.16 are 0.8 and 0.2. Since p < q, p = 0.2 and q = 0.8. The frequency of the recessive phenotype is $q^2 = (0.8)^2 = 0.64$. The total number of recessive individuals = 0.64 × 2000 = 1280.

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139. (3)

The major coordinating centre is the thalamus. The structure located at its base is the hypothalamus. The hypothalamus controls body temperature, urge for eating, and drinking, and contains neurosecretory cells.

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140. (2)

List-I	List-II
Ctenoplana	Exclusively marine and shows bioluminescence.
Pinctada	Possesses a soft and spongy mantle over a visceral hump.
Taenia	Dorso-ventrally flattened body
Locusta	Gregarious pest with a chitinous exoskeleton.

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141. (2)

A fall in GFR activates JG cells to release renin. Renin converts the plasma protein angiotensinogen into angiotensin I, which is further converted to the powerful vasoconstrictor angiotensin II.

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142. (2)

Ribozymes are RNA enzymes, not proteins. Some enzymes (e.g., thermophilic) work above 40°C. ES complex formation is essential in enzyme action. Nucleotides have a phosphate group, nucleosides don't. Palmitic acid is a 16-carbon saturated fatty acid.

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143. (1)

The gland in question is the thymus, which degenerates in old individuals resulting in decreased production of thymosins and a weakened immune response. It is located between the lungs behind the sternum on the ventral side of the aorta.

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144. (1)

List-I	List-II
FRC	ERV + RV
IC	TV + IRV
EC	TV + ERV
TLC	RV + ERV + TV + IRV

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145. (1)

Placenta provides a vascular connection between the foetus and the maternal tissues (uterus) for exchange of gases, nutrients, and wastes.

The zygote undergoes cleavage to form a blastocyst, which then implants in the uterus.

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146. (2)

Multiload 375 is a copper-releasing Intra Uterine Device (IUD). IUDs are inserted by doctors/nurses in the uterus through the vagina, and they fundamentally increase the phagocytosis of sperms within the uterus.

(NCERT Ed. 2025-26, 12th, Page No. 44)

147. (3)

Plasmodium requires two hosts (human and *Anopheles* mosquito) to complete its life cycle. When it ruptures the red blood cells, it releases haemozoin, a toxin responsible for the recurring chill and high fever of malaria.

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[Contd...

148. (4)

It produces inactive protoxins (Cry proteins) which become active in the alkaline pH of insect gut, leading to cell lysis.

These genes code for Bt toxins specific to cotton bollworms, and are widely used in Bt cotton.

Rosie was the first transgenic cow that produced human alpha-lactalbumin-enriched milk, making it nutritionally improved

RNA interference (RNAi) takes place in all eukaryotic organisms as a method of cellular defense, not prokaryotic organisms.

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149. (3)

According to the text, utilizing the energy from ATP hydrolysis, the myosin head binds to the exposed active sites on actin to form a cross-bridge. This cross-bridge physically pulls the attached actin filaments towards the centre of the 'A' band.

(NCERT Ed. 2025-26, 11th, Page No. 222)

150. (3)

A cockroach can survive for about one week without its head because basic life processes like respiration occur via spiracles and tracheae, not controlled by the head.

Death eventually occurs due to starvation and dehydration, not immediate failure of vital systems.

The rest of the nervous system is situated along the ventral (belly-side) part of its body, not the dorsal part.

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151. (4)

The primary structure of a protein refers specifically to the sequence of amino acids and the positional information (N-terminal to C-terminal) within the polypeptide chain.

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152. (2)

If both parents are Rh-negative, their genotype must be dd. Since they can pass only the d allele, all their children will also be dd (Rh-negative). Therefore, an Rh-positive child is not possible. Rh-positive parents can have genotypes DD or Dd. If both parents are heterozygous (Dd × Dd), there is a chance that a child receives d from both parents, resulting in dd (Rh-negative). Hence, Rh-positive parents can have an Rh-negative child.

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153. (2)

About 2000 million years ago (mya) the first cellular forms of life appeared on earth.

Homo erectus had a brain size around 900 cc and is believed to be eating meat.

Neanderthals lived in Europe and Central Asia, not East Africa.

Australopithecines were primarily fruit eaters and did not use stone tools for hunting (tool use seen in later Homo).

Dryopithecus was more ape-like and *Ramapithecus* more man-like.

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154. (2)

The Loop of Henle plays a significant role in maintaining the high osmolarity of the medullary interstitial fluid through the counter-current mechanism, even though reabsorption is minimum in its ascending limb.

(NCERT Ed. 2025-26, 11th, Page No. 209)

155. (2)

The cranial meninges consist of an outer dura mater, a very thin middle layer called the arachnoid, and an inner layer called the pia mater.

(NCERT Ed. 2025-26, 11th, Page No. 235)

156. (4)

Fertilization in humans typically occurs in the Ampulla, the widest part of the oviduct, where the sperm meets the ovum.

(NCERT Ed. 2025-26, 12th, Page No. 35)

157. (1)

Graph (i) - Thermoregulation: Line X maintains a constant internal temperature despite external changes (Regulators - mammals and birds).

Line 'Y's internal temperature matches the external environment (Conformers - reptiles, amphibians).

Graph (ii) Body Size rule: Type a has a high Surface Area to Volume ratio, meaning they are very small animals. Type b has a low SA/V ratio, meaning they are large animals.

Humming bird: Bird (Regulator = X), Small size (Type a) $\rightarrow X_a$

Crocodile: Reptile (Conformer = Y), Large size (Type b) $\rightarrow Y_b$

Frog: Amphibian (Conformer = Y), Small size (Type a) $\rightarrow Y_a$

Polar bear: Mammal (Regulator = X), Large size (Type b) $\rightarrow X_b$

The correct sequence is X_a; Y_b; Y_a; X_b.

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158. (4)

DNA fragments cannot be seen in visible light without staining. They must be stained with ethidium bromide and exposed to UV radiation. A protein produced by expressing a gene in a heterologous host is called a recombinant protein.

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159. (2)

Systemic venous blood has $pO_2 \approx 40$ mm Hg due to oxygen usage by tissues.

Systemic arterial blood has $pO_2 \approx 95$ mm Hg after oxygenation in lungs.

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160. (1)

Glucagon is called a hyperglycemic hormone because it raises blood glucose levels when they fall below normal. It acts opposite to insulin and helps maintain glucose homeostasis.

Glucagon acts mainly on hepatocytes (liver cells) to stimulate glycogenolysis (breakdown of glycogen) and gluconeogenesis (formation of glucose). Both processes increase glucose release into blood.

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161. (3)

Cocaine (coke or crack) is obtained from the coca plant (*Erythroxylum coca*) native to South America. It is a powerful CNS stimulant, producing euphoria and increased energy. Heroin is a depressant. Cannabinoids mainly affect the cardiovascular system.

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162. (3)

Red muscle fibres appear reddish due to high myoglobin content, which stores oxygen.

They are adapted for sustained, aerobic activities (e.g., posture, endurance).

White muscle fibres mainly depend on anaerobic glycolysis, not aerobic processes.

They are suited for quick, powerful contractions and fatigue rapidly.

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163. (3)

Increased incidences of concealed (hidden/untreated) STIs due to social stigma is a major threat to reproductive health. Improved health is indicated by better detection and cure of STIs, not their concealment.

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164. (3)

List-I	List-II
Gene therapy	Insertion of genes into an individual's cells and tissues to treat diseases
Golden Rice	A genetically modified crop enhanced with Vitamin 'A'.
Bacillus thuringiensis	Produces a toxic insecticidal crystalline protein.
Somatic hybridization	Isolation and fusion of naked protoplasts from two different varieties of plants.

(NCERT Ed. 2025-26, 12th, Page No. D179)

165. (2)

The crop is followed by the gizzard or proventriculus. It has an outer layer of thick circular muscles and a thick inner cuticle forming six highly chitinous plates called teeth, which help in grinding food.

(NCERT Ed. 2020-21, 11th, Page No. 113)

166. (3)

Many bony fishes, aquatic amphibians, and aquatic insects excrete ammonia and are called ammonotelic. Mammals and marine fishes are ureotelic, while reptiles, birds, and land snails are uricotelic.

(NCERT Ed. 2025-26, 11th, Page No. 205)

167. (2)

A closer examination reveals that the relative abundance of carbon and hydrogen with respect to other elements is higher in any living organism than in earth's crust.

(NCERT Ed. 2025-26, 11th, Page No. 104)

168. (3)

The coronary system of blood vessels is present exclusively for the circulation of blood to and from the cardiac musculature (the heart tissue itself).

(NCERT Ed. 2025-26, 11th, Page No. 202)

169. (2)

When a stimulus is applied, the membrane becomes freely permeable to Na^+, leading to a rapid influx of Na^+ which reverses the polarity, causing depolarisation.

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[Contd...

170. (4)

Interferons are antiviral proteins released by virus-infected cells. They do not kill viruses directly, but protect neighboring uninfected cells. They induce antiviral state → inhibit viral replication. Part of innate immunity.

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171. (4)

List-I	List-II
Homologous organs	Forelimbs of whales, bats, cheetahs, and humans
Analogous organs	Sweet potato (root) and potato (stem)
Adaptive radiation	Darwin's finches in the Galapagos Islands
Convergent evolution	Placental wolf and Tasmanian wolf

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172. (3)

The scrotum maintains the temperature 2-2.5°C lower than normal body temperature. Leydig cells are present in the interstitial spaces between seminiferous tubules. They secrete androgens (mainly testosterone).

The epididymis leads into vas deferens, which ascends and loops over the urinary bladder.

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173. (2)

Bacterial cells are made competent by treating them with a specific concentration of a divalent cation, such as calcium, which increases the efficiency with which DNA enters the bacterium through pores in its cell wall.

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174. (4)

The adrenal cortex has three layers:

Zona glomerulosa → secretes mineralocorticoids like aldosterone

Zona fasciculata → secretes glucocorticoids

Zona reticularis → secretes androgens

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175. (2)

List-I	List-II
Morphine	*Papaver somniferum*
Cocaine	*Erythroxylum coca*
Cannabinoids	*Cannabis sativa*
Penicillin (Antibiotic context)	Fungi / Microbes

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176. (2)

Asthma is characterized by difficulty in breathing causing wheezing due to inflammation of the bronchi and bronchioles. Emphysema involves alveolar damage. Occupational disorders involve fibrosis.

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177. (3)

In cartilaginous joints, bones are joined by cartilage. The joint between adjacent vertebrae in the vertebral column is of this pattern, permitting limited movement. Skull bones are fibrous joints. Humerus/pectoral and carpal/metacarpal are synovial joints.

(NCERT Ed. 2025-26, 11th, Page No. 227)

178. (3)

Ovulation occurs around day 14, and sperms can survive 2–3 days. Hence, this window is the most fertile period.

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179. (1)

The Ti plasmid of *Agrobacterium tumefaciens* has been modified (disarmed) to remove tumor-causing genes. It is now used as a highly effective cloning vector for plant genetic engineering.

Agrobacterium tumefaciens is a plant pathogen (mainly dicots) that naturally transfers T-DNA into plant cells, causing tumors (crown gall).

This natural gene transfer ability is the basis for its use in biotechnology.

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180. (3)

Gap junctions facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells, allowing the rapid transfer of ions and small molecules. Tight junctions stop leakage, and adhering junctions perform cementing.

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